



DATA SCIENCE



Engr. Dr. Muhammad Nadeem Majeed



Today's Agenda

- About Myself
- Course Information and Protocols
- Data Data Everywhere
- Categories of Data
- What is Data Science?
- Factors making Data Science Ubiquitous
- Applications of Data Science
- Discussion on Course Matrix



Engr. Dr. Muhammad Nadeem Majeed

Associate Professor
Department of Data Science

- PhD Computer Engineering
- MS Computer Engineering

Certifications:

- 1. Project Management Professional (PMP)®**
- 2. PRINCE2 Agile Practitioner**
- 3. Professional Scrum Master (PSM)**
- 4. Certified Lean Six Sigma Green Belt (CSSC)**
- 5. ITIL Certified**
- 6. Cisco Certified Network Associate (CCNA)**
- 7. Juniper Networks Certified Internet Associate (JNCIA)**
- 8. IBM Data Engineer**

Research interests:

Mispronunciation Detection,
Disease classification.
Network and Communications.
Machine Learning

Experience: 22 Years of Teaching & Network Management

- University of The Punjab, Lahore
- University of Engineering and Technology Taxila.
- NUST School of Electrical Engineering and Computer Science. Islamabad.
- University of Arid Agriculture Rawalpindi.



Course Information and Protocols



Course Info

- **Textbook(s):** Python for Data Analysis, by, Wes McKinney, 2nd Edition, Published in 2017, ISBN-13: 9781491957660
- **Lectures Slides Available at: Goole Class Room**
- **Codes Hosted at:** <https://github.com/nadeem-majeedch/DataScienceJourney>
- **Grades Website:** <http://online.pucit.edu.pk>
- **Prerequisites: Basic Programming skills**
- **Office:** Room #37, Building-A, FCIT (NC)
- **Students Counseling hours:**
 - Mon: 1500 hrs – 1600 hrs
 - Wed: 1500 hrs – 1600 hrs
- **24 hour turnaround for email:** nadeem.majeed@pucit.edu.pk



Cheating Policy

- Academic integrity
- Both the cheater and the student who aided the cheater will be held responsible for the cheating
- The instructor may take actions such as:
 - require repetition of the subject work,
 - assign 'zero' or may be 'negative' marks for the subject work,
 - for serious offenses, assign an **F** grade for the **course**





Late Policy for Home Works and PA

- Late policy for Assignment, Quizzes, and other deliverables
 - No late Assignment submissions!
 - No late quizzes or exams!
- Sticking to dates is your responsibility!
 - Check announcements on lecture notes regularly
- Your best strategy is to play it safe – submit everything on time



Lecture Format

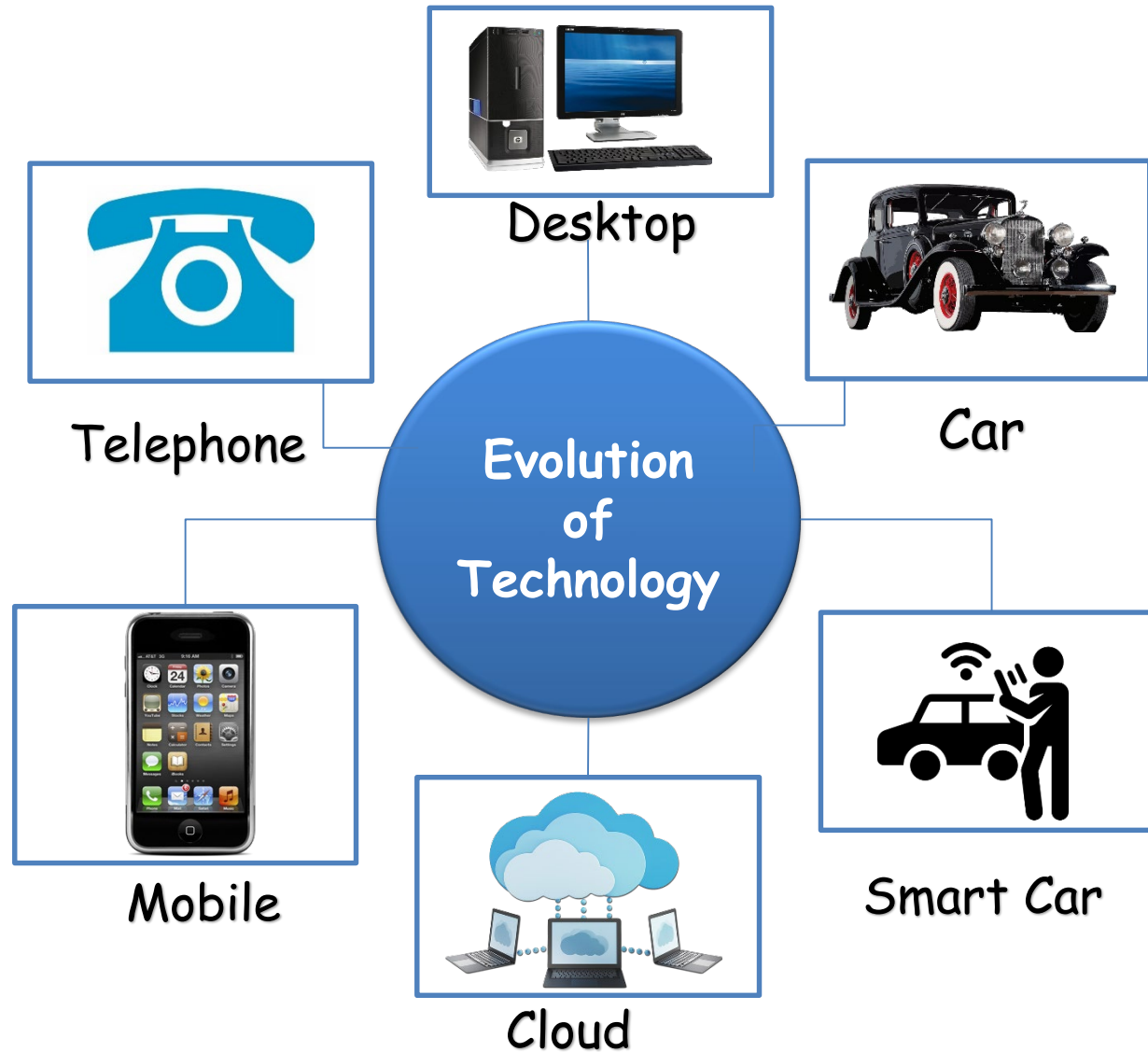




Data Data Everywhere Data Sources



Data Sources: Evolution of Technology





Instructor: Dr. Muhammad Nadeem Majeed



Data Sources: Social Media



347,222 tweets



1,736,111 pictures



4,166,667 likes &
200,000 photos



204,000,000 emails



YouTube

300 hrs of video uploaded
2.78 million video views



342,000 apps downloaded



70,017 hours watched



2.4 million search queries

Imagine processing & analyzing this much data, and then trying to figure out important insights from it



Data Sources: Other Factors



Data Science is all about extracting the useful insights from data and using it to grow your business



Categories of Data



Structured Data

Examples


Social Security
Number


Date


Phone
800 000
Phone Numbers


Customer
Name


Transaction
information


Credit Card
number

**Structured
Data**

Applications



Airline reservation
systems



Inventory control



Pre-defined
Data Model



Easy to Search



Text-based

Characteristics



Data Mart

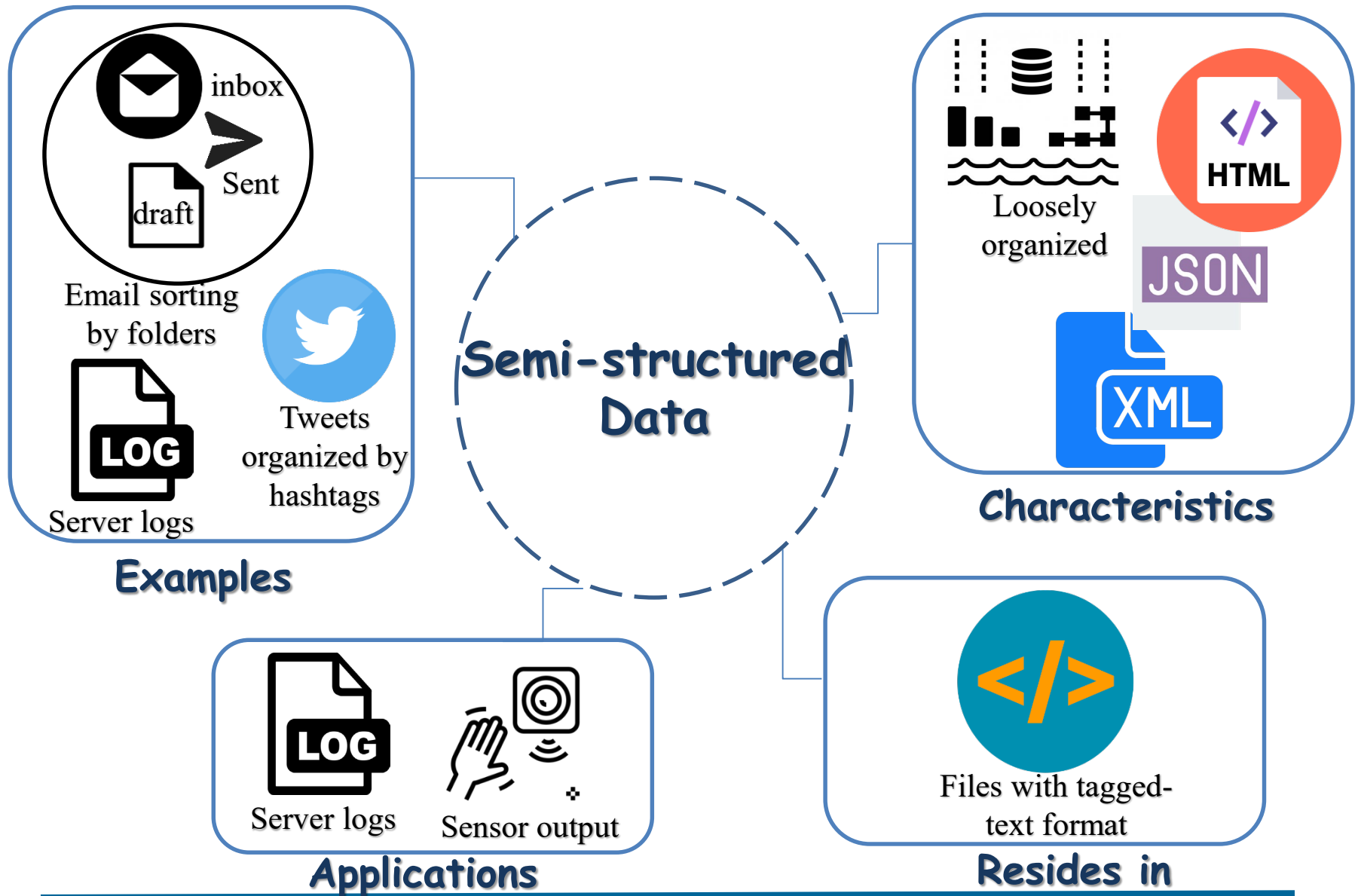


Database

Resides in

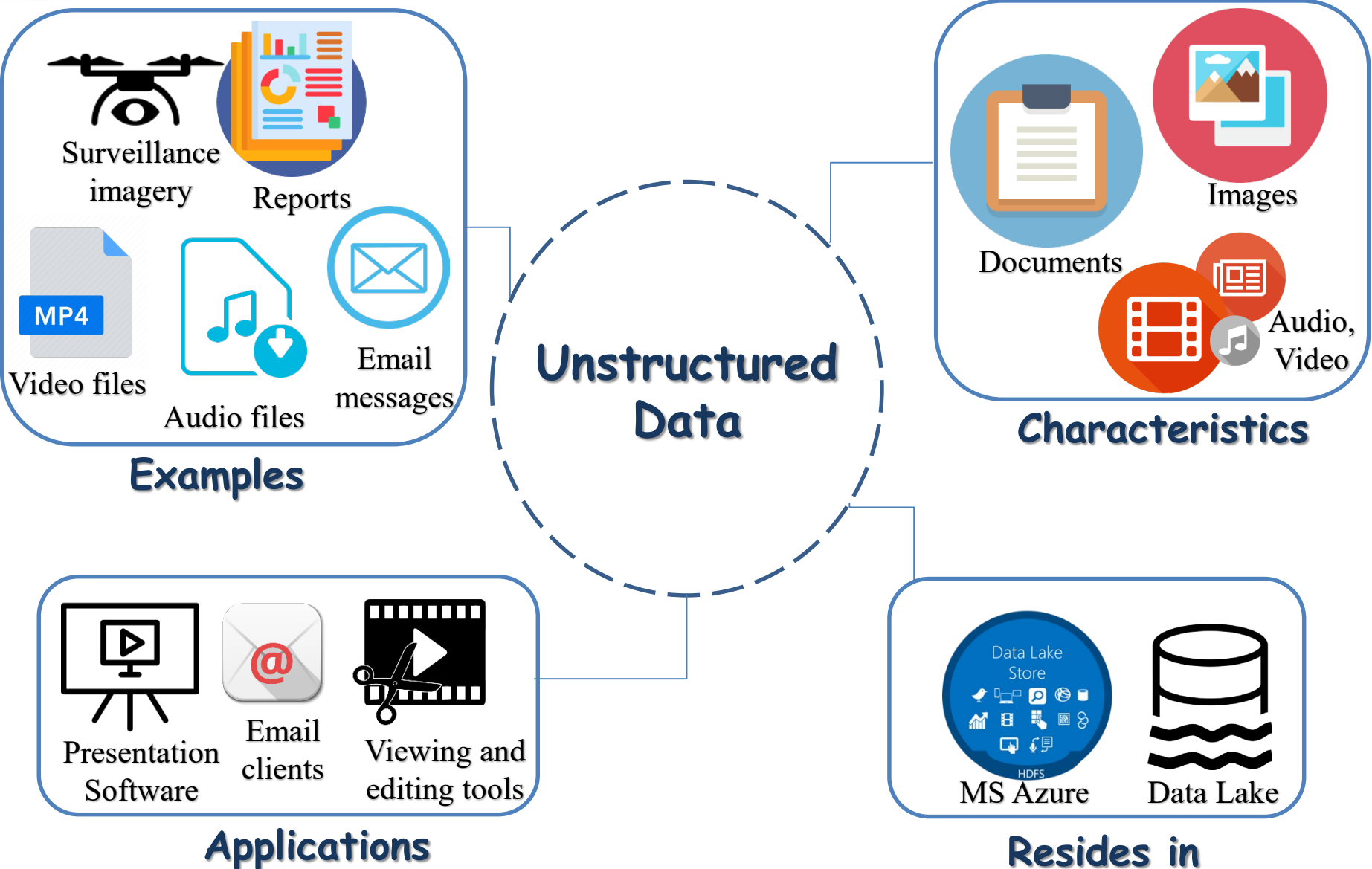


Semi-structured Data



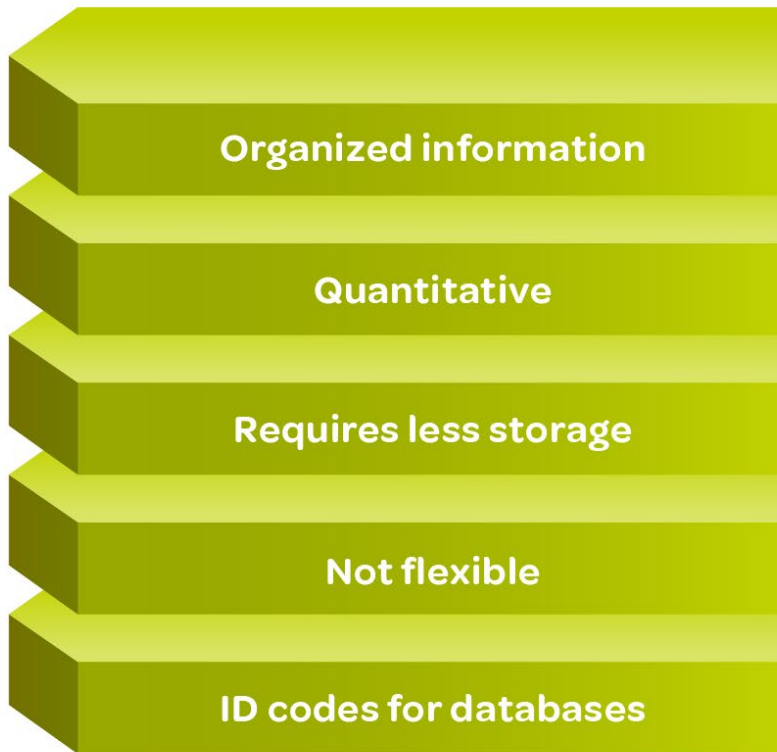


Unstructured Data

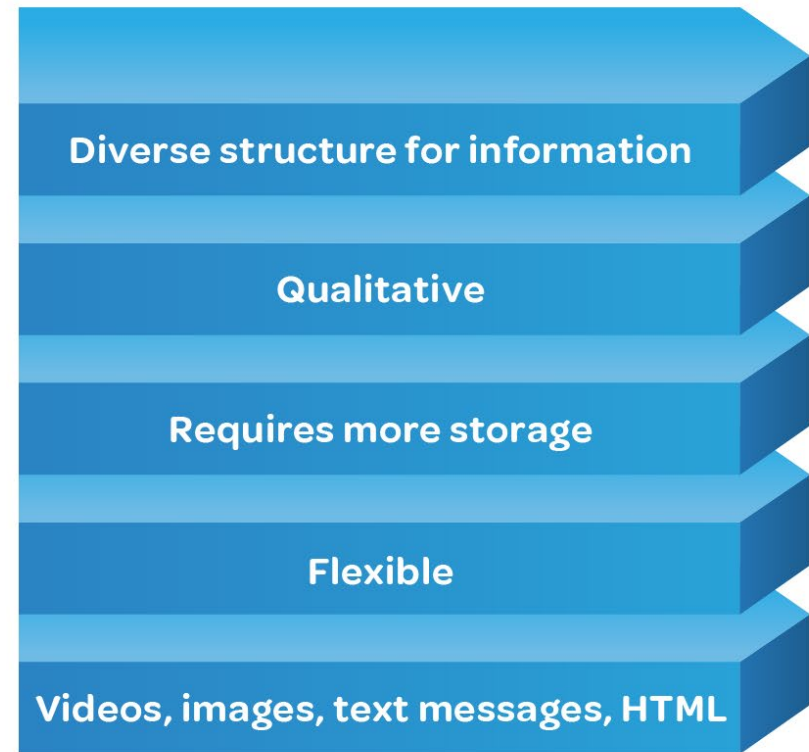


Key Differences Between Structured and Unstructured Data

STRUCTURED DATA



UNSTRUCTURED DATA



Typical human-generated unstructured data includes:

Text Files: Word processing, spreadsheets, presentations, emails, logs.

Email: Email has some internal structure thanks to its metadata, and we sometimes refer to it as semi-structured. However, its message held is unstructured and traditional analytics tools cannot parse it.

Social Media: Data from Facebook, Twitter, LinkedIn, YouTube, Instagram, photo sharing sites.

Mobile data: Text messages, locations.

Communications: Chat, IM, phone recordings, collaboration software. **Media:** MP3, digital photos, audio and videos.

Business applications: MS Office documents, productivity applications.

Typical machine-generated unstructured data includes:

Satellite imagery: Weather data, landforms, military movements.

Scientific data: Oil and gas exploration, space exploration, seismic imagery, atmospheric data. **Digital surveillance:** Surveillance photos and video.

Sensor data: Traffic, weather, oceanographic sensors.

Database vs Data Marts vs Data warehouse vs Data Lake

Database: A database is an electronic repository for structured data from a single source where you can store, retrieve, and query it for a specific purpose.

Data marts: are databases that hold a limited amount of **structured data for one purpose** in a single line of business.

Here is an example. A data mart can be a database of organized data for your sales and marketing department that does not exceed 100 Gigabytes (GB).

Data warehouse is a relational database that can handle, store, and bring to one place structured data sets **coming from multiple sources**. Data warehousing supports business decision-making by analyzing varied data sources and reporting them in an informational format. Snowflake and Amazon Redshift

Data Lake is a large repository that houses **structured, semi-structured, and unstructured data from multiple sources**. Data lakes are also an excellent feeding ground for big data, artificial intelligence, and machine learning programs. However, data lakes can be tough to derive insights for everyday business needs unless you are a data specialist. This is where other types of standardized data storing options come in. **Amazon Web Services Data Lake, Microsoft Azure Data Lake Storage, Intelligent Data Lake by Informatica**

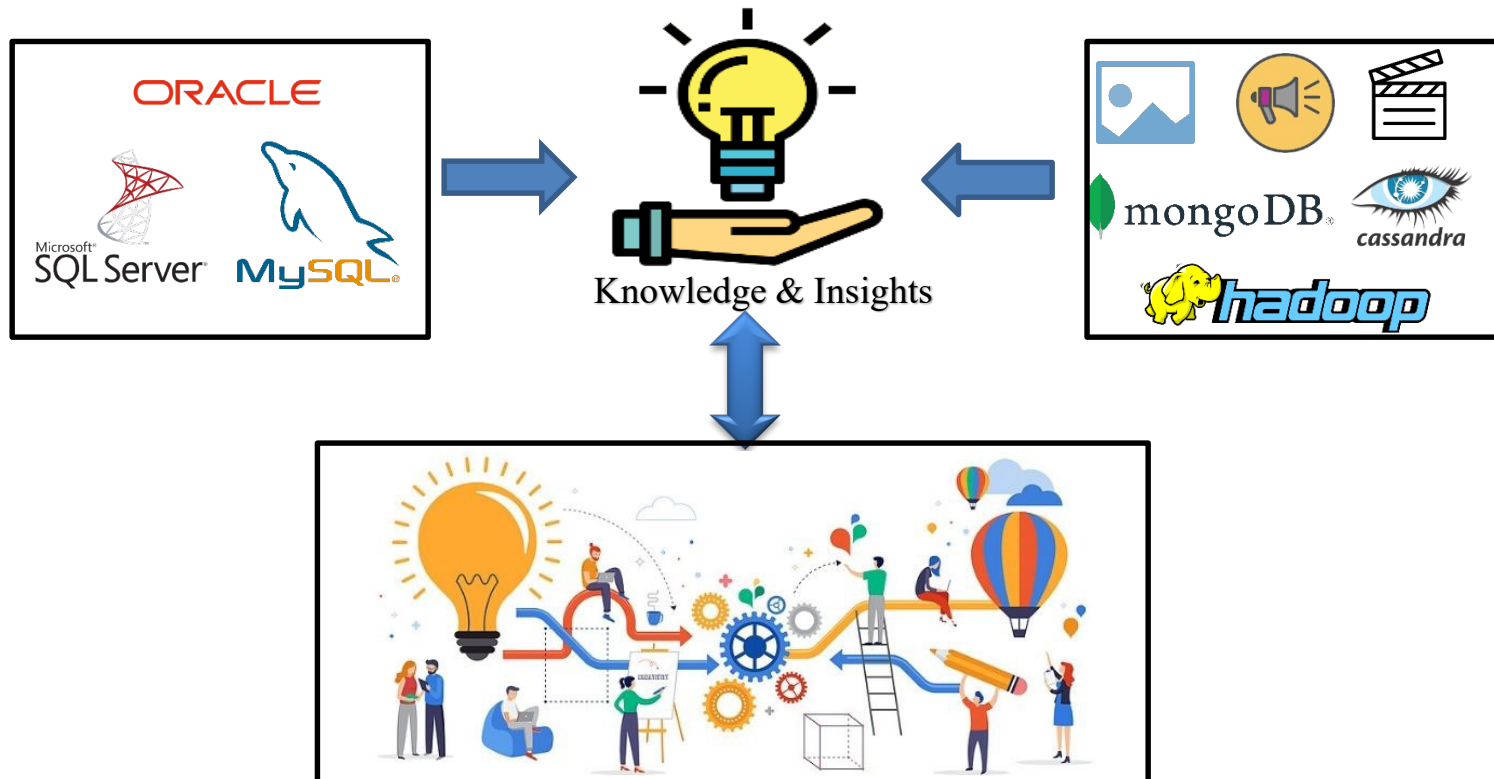


What is Data Science?



What is Data Science?

Data Science is an Inter-Disciplinary Field that uses





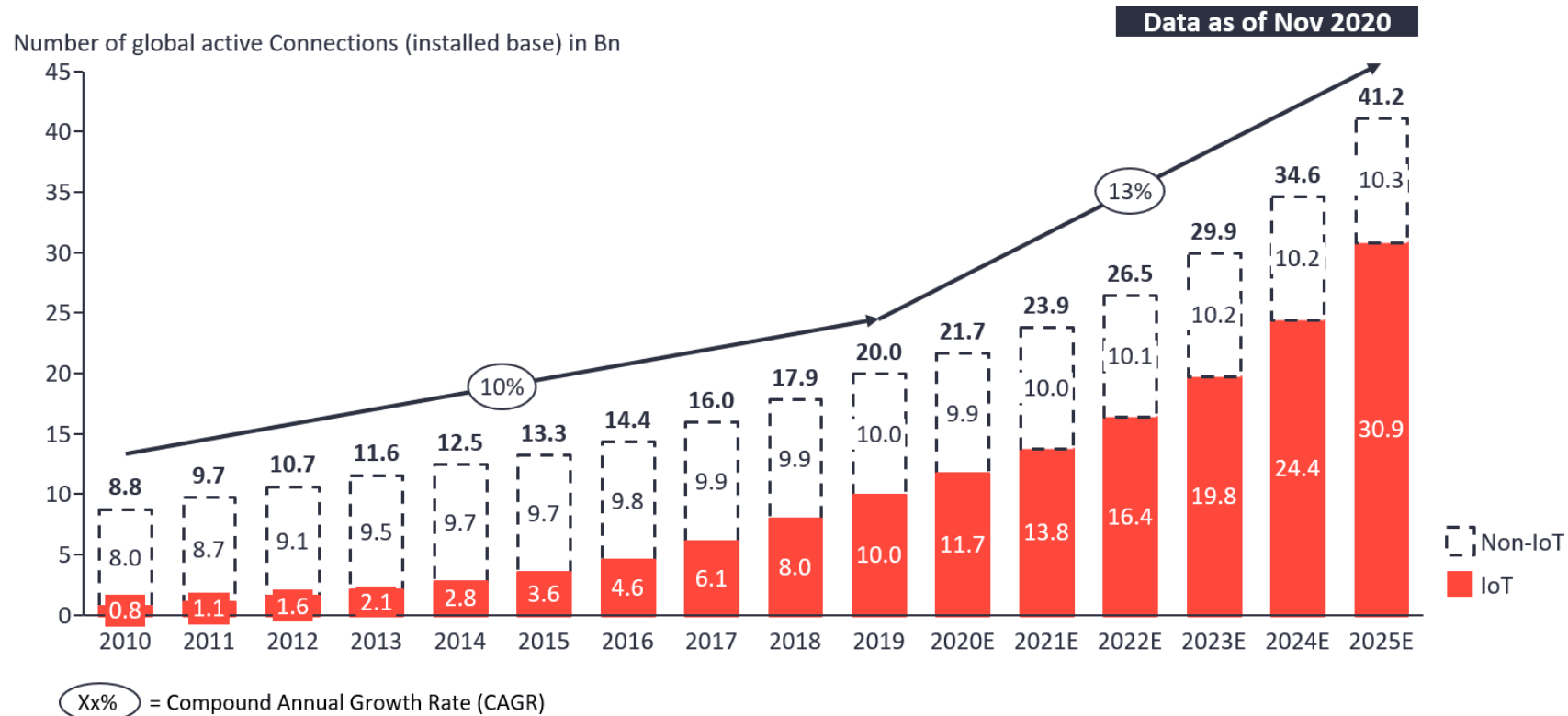
Factors Making Data Science Ubiquitous



Increasing Number of Connected Devices

Total number of device connections (incl. Non-IoT)

20.0Bn in 2019– expected to grow 13% to 41.2Bn in 2025

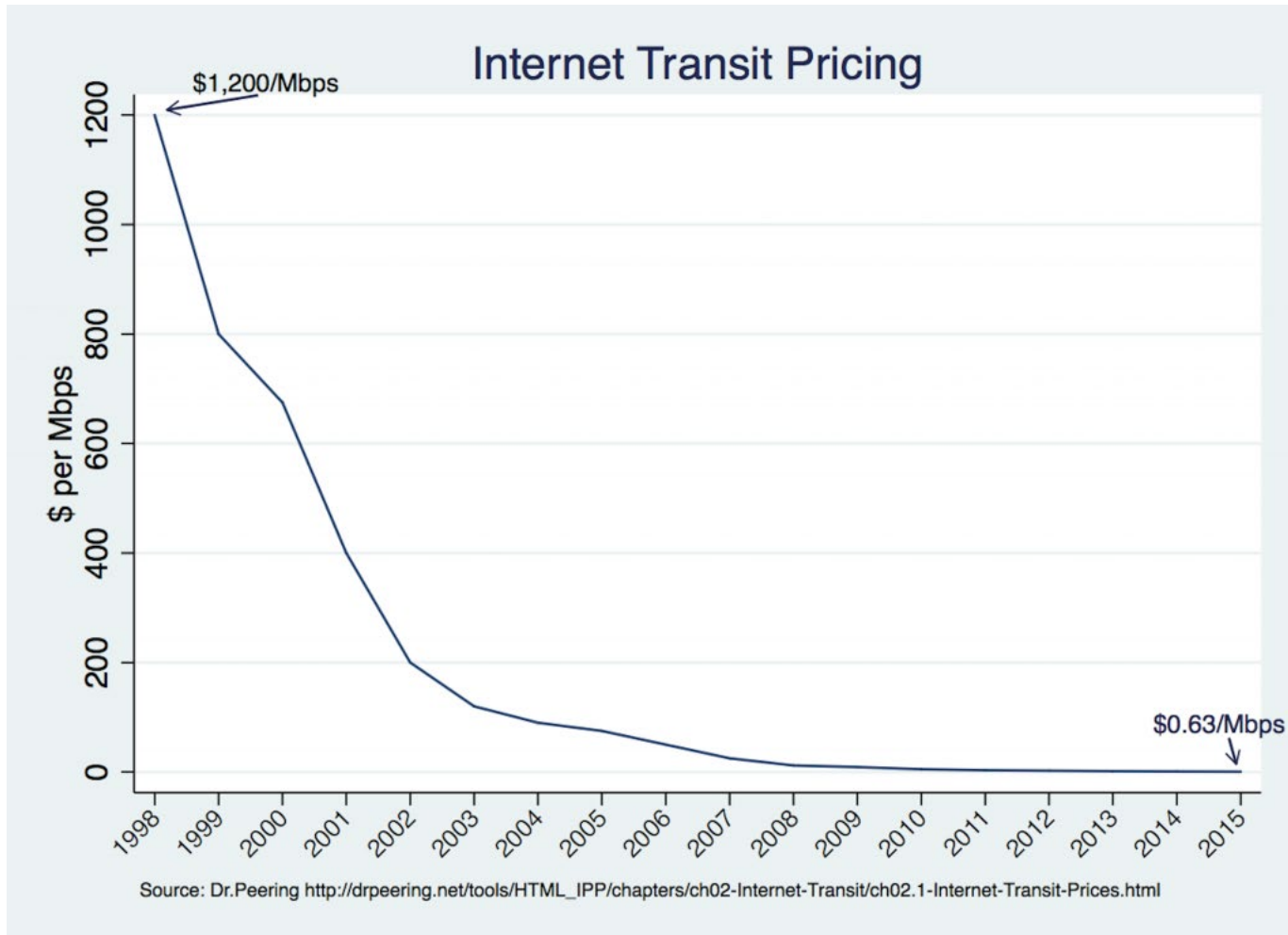


Note: Non-IoT includes all mobile phones, tablets, PCs, laptops, and fixed line phones. IoT includes all consumer and B2B devices connected – see IoT break-down for further details

Source(s): IoT Analytics - Cellular IoT & LPWA Connectivity Market Tracker 2010-25

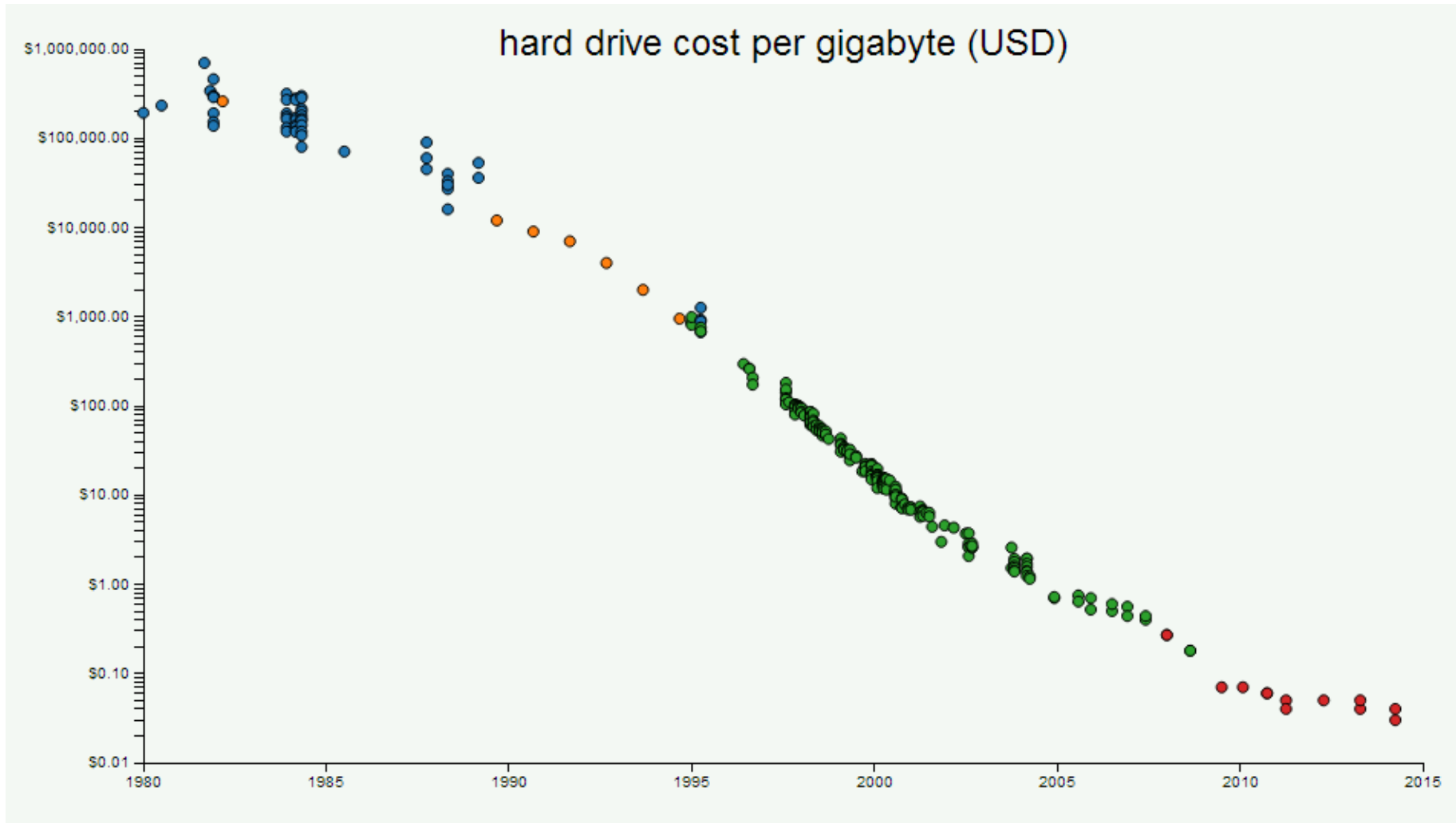


Decreasing Internet Transit Pricing





Decreasing Costs for Data Storage



Source: <https://community.spiceworks.com>



Decreasing Computational Costs

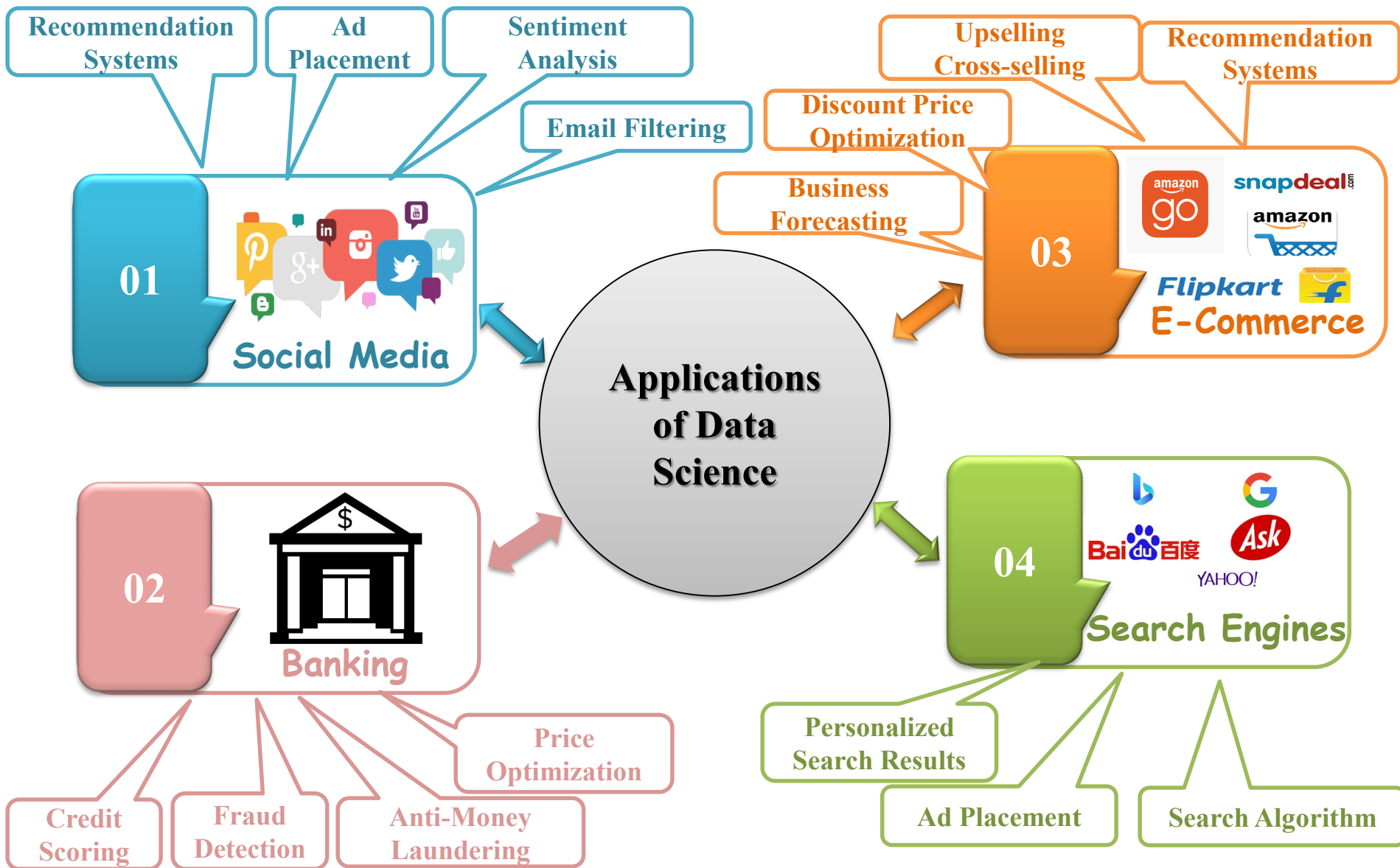




Applications of Data Science?

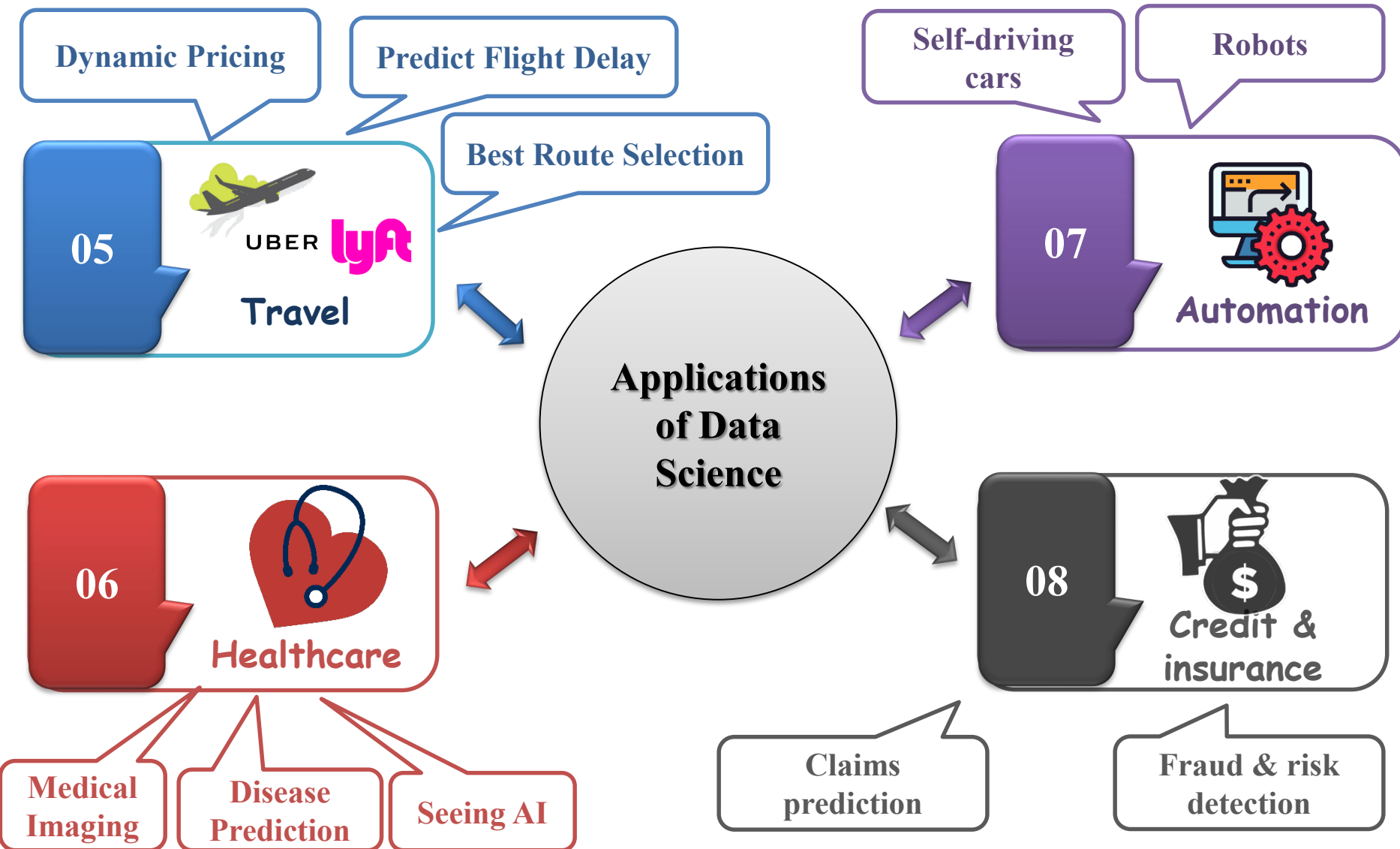


Applications of Data Science





Applications of Data Science (cont...)





Discussion on Course Matrix



What we will do in this course?

Module 1: (Overview of the course)

- What is Data Science?
- Why/How to do Data Science?
- Structured vs Unstructured data
- Applications of Data Science
- Tools and Technologies for Data Science
- Life Cycle of a Data Science Project
- Job Roles in the Industry
- Data Science Use Cases from real life
- Git and Github for Data Scientists

Reading Tasks:

- ...



What we will do in this course?

Module 2: (Basics of Python Programming)

- Overview of Python programming language
- Python programming environments
- Python intrinsic data types and operators
- Python data structures
- Selection and Repetition structures
- Functions in Python
- Exception handling
- Modules, packages and libraries
- Basic file handling in Python

Reading Tasks:

- ...



What we will do in this course?

Module 3: (Python for Data Scientists)

- Overview of Python libraries for Data Science
- Reading data in Python (csv, xlsx, json)
- Data manipulation with NumPy
- Scientific computation with SciPy
- Data manipulation with Pandas
- Visualization with Matplotlib and Seaborn

Reading Tasks:

- ...



What we will do in this course?

Module 4: (Mathematics for Data Scientists)

- Applied Linear Algebra for Data Scientists
- Applied Calculus for Data Scientists
- Applied Descriptive Statistics for Data Scientists
- Applied Inferential Statistics for Data Scientists

Reading Tasks:

- ...



What we will do in this course?

Module 5: (Data Acquisition)

- Overview of Data Acquisition
- Data Acquisition from SQL Databases
- Data Acquisition from NoSQL Databases
- Data Acquisition from Websites

Reading Tasks:

- ...



What we will do in this course?

Module 6: (Data Wrangling and EDA)

- Acquire data sets from different sources
- Understand the datasets
- Transform to appropriate format
- Perform data cleaning
- Perform data wrangling
- Carry out Exploratory Data Analysis
- Identify pattern/trends using different visualization and statistical tools
- Preparing the dataset ready to be used by Machine Learning Engineer

Reading Tasks:

- ...



What we will do in this course?

Module 7: (Machine Learning)

- Overview of Machine Learning
- Categories of Machine Learning Types and Algorithms
- Python for Machine Learning (Scikit-learn)
- Will do hands on practice for
 - ✓ Model creation
 - ✓ Model training
 - ✓ Model evaluation
 - ✓ Feature engineering
 - ✓ Dimensionality reduction

Reading Tasks:

- ...



What we will do in this course?

Module 8: (Deep Learning: A Bird's-eye View)

- Machine Learning vs Deep Learning
- Overview of Deep Learning Models (CNN vs RNN)
- Deep Learning Applications
 - ✓ Natural Language Processing
 - ✓ Image recognition
 - ✓ Self-driving cars
 - ✓ Language translation services
- A Hello World on Deep Learning Project using
 - ✓ TensorFlow/Keras/Theano/Torch/Caffe

Reading Tasks:

- ...



What we will do in this course?

Module 9: (Big Data: A Bird's-eye View)

- What is Big Data?
- Big Data Storage and Processing Frameworks
 - ✓ Apache Hadoop with MapReduce (used by Alibaba, AOL)
 - ✓ Apache Storm (used by Twitter, Spotify)
 - ✓ Apache Spark (used by Netflix, Yahoo, eBay)
 - ✓ Apache Hive (used by Facebook, Walmart)
- An Overview of Hadoop Ecosystem
 - ✓ Data Storage (HDFS, HBASE)
 - ✓ Data Processing (YARN, Map Reduce)
 - ✓ Data Access (Hive, Pig, Mahout, Avro, Sqoop)
 - ✓ Data Management (Oozie, Chukwa, Flume, ZooKeeper)

Reading Tasks:

- ...



Things To Do

- Memorize and follow class protocols
- Should have a very clear understanding of different data sources, its types and storage
- Must know the applications of data science in different domains
- Visit the resource web-sites and download all uploaded resources
- While going through today's lecture slides click all the tools and technologies, which have been hyperlinked to respective web sites.
- You should be able to give a single line description of each tool and technology mentioned in lecture slides



Coming to office hours does NOT mean you are academically weak!